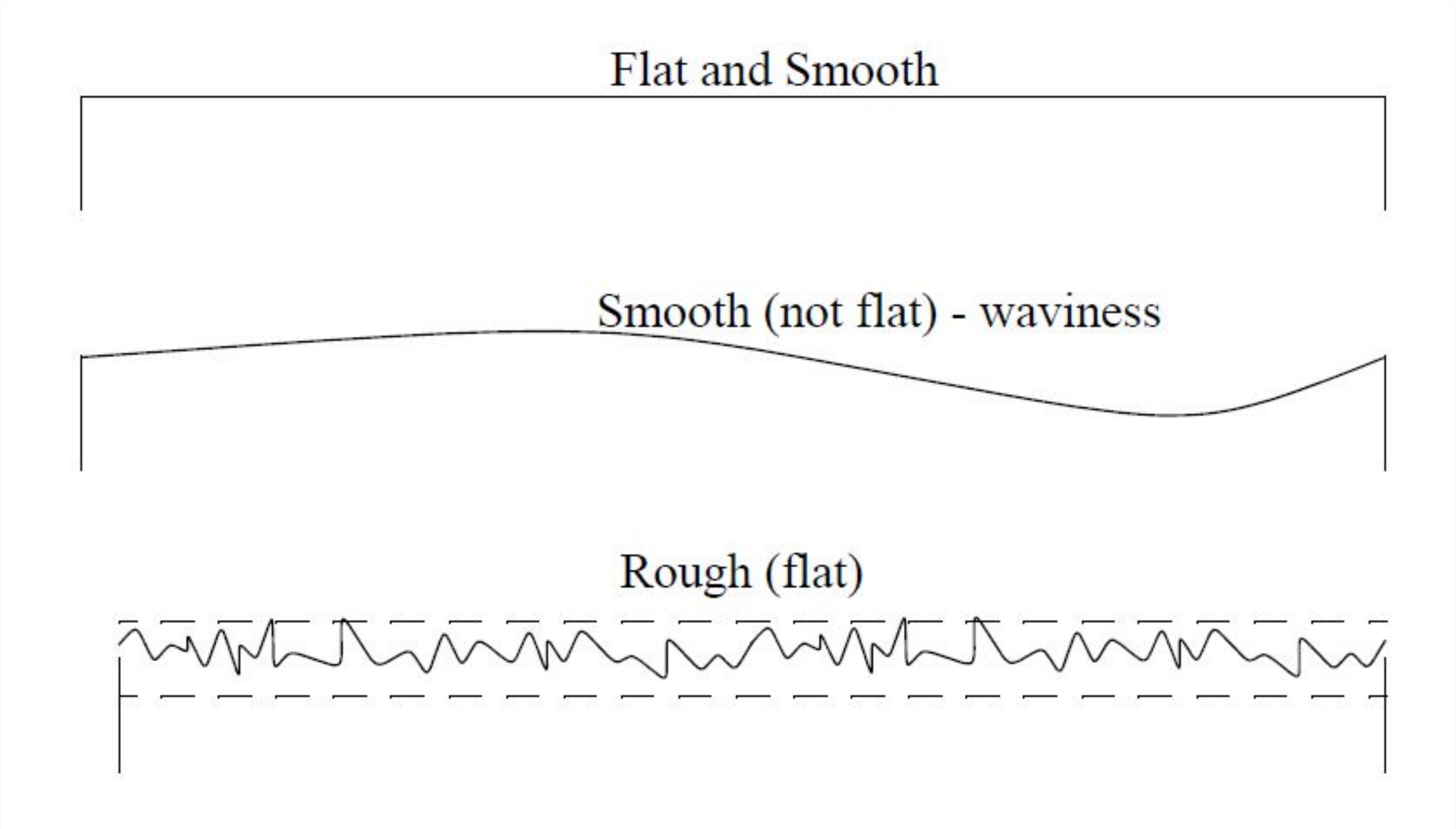
SURFACE GEOMETRY

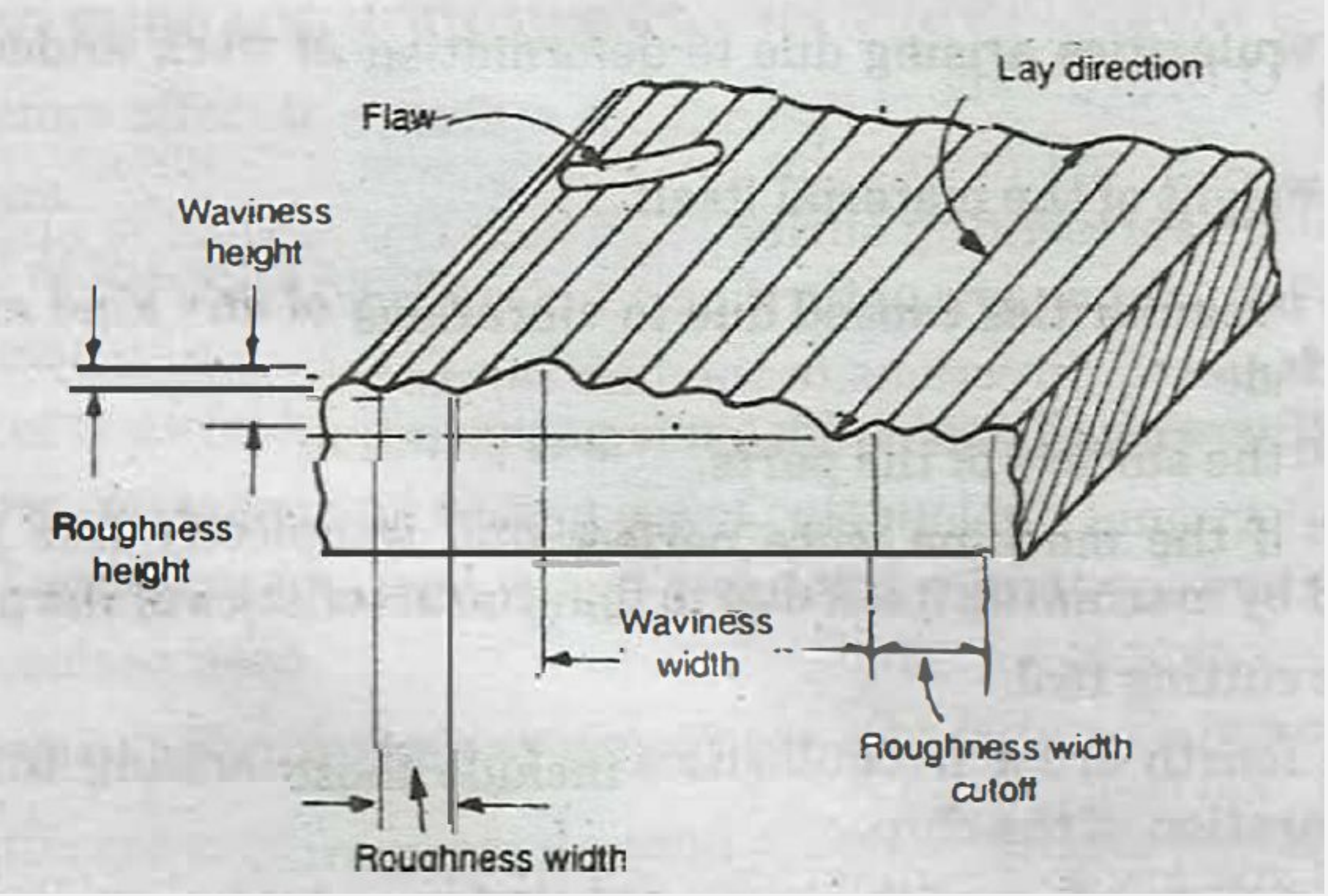
BEARING AREA CURVE

The bearing area curve is also called as **Abbot's bearing curve**. This is determined by adding the lengths a, b, c etc. at depths x, y, z etc. below the reference line and indicates the percentage bearing area which becomes available as the crest area worn away.



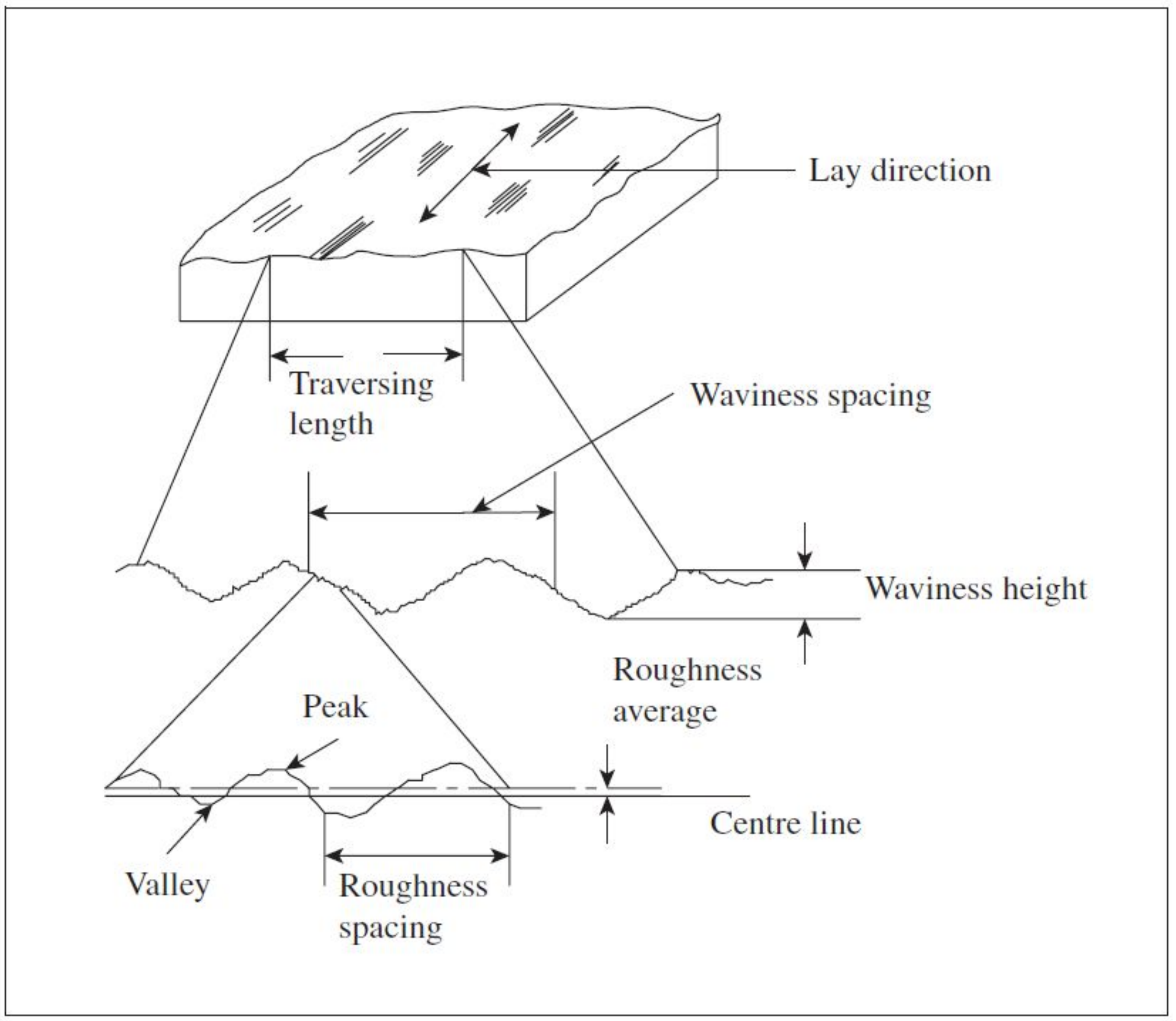
Real surfaces are rarely so flat, or smooth, but most commonly a combination of the two

SURFACE TOPOGRAPHY



Lay direction
Flaw
Waviness
height
Roughness
height
Waviness
width
Roughness width
cutoff
Roughness width

WAVINESS AND ROUGHNESS



FACTORS AFFECTING SURFACE ROUGHNESS

(1) Vibrations

(2) Material of the workpiece

(3) Type of machining

(4) Rigidity of the system consisting of machine tool, fixture cutting tool and work

(5) Type, form, material and sharpness of cutting tool

(6) Cutting conditions *i.e.,* feed, speed and depth of cut

(7) Type of coolant used

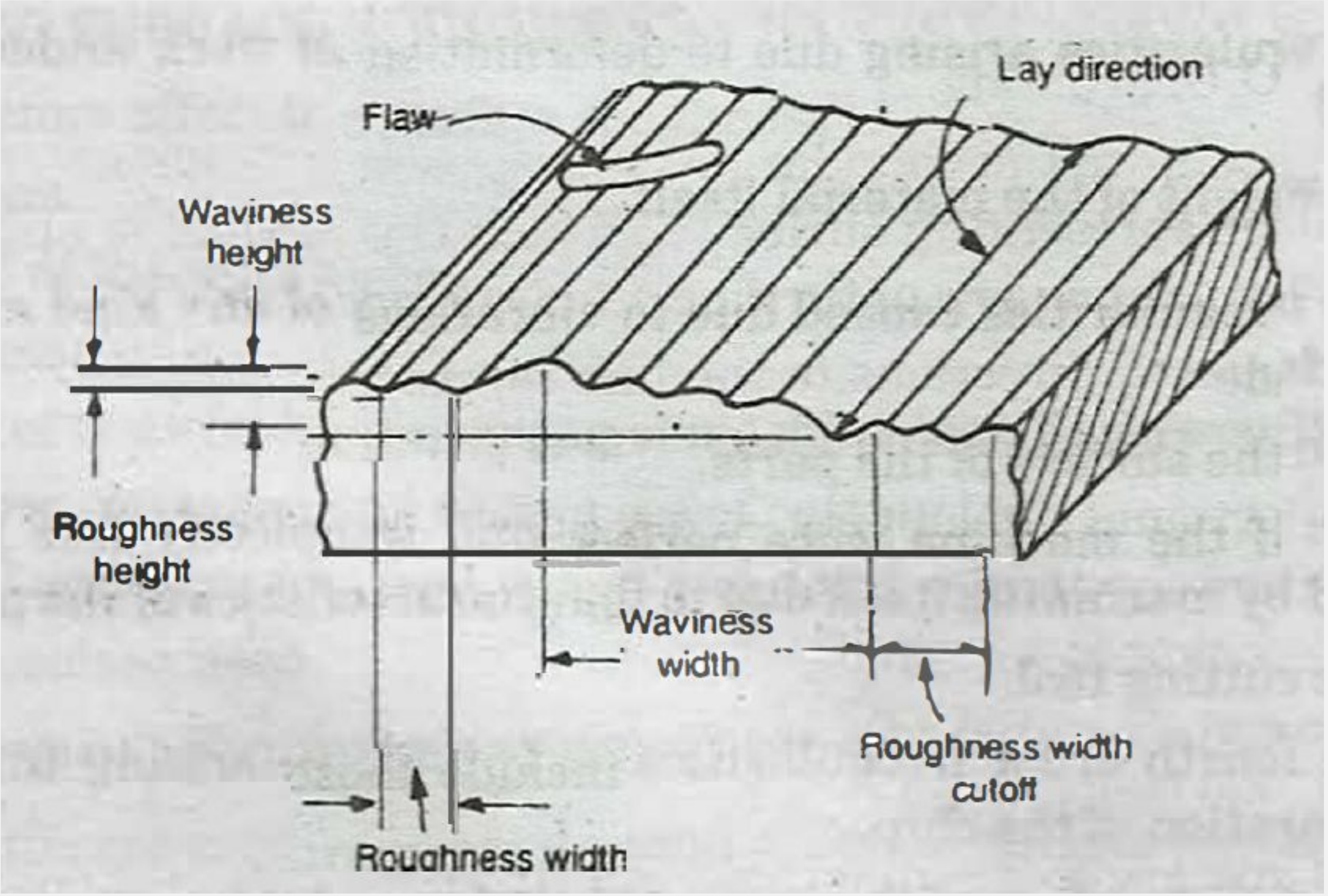
(I) PRIMARY TEXTURE (ROUGHNESS)

- **The surface irregularities of small wavelength are called primary texture or roughness.**
- **These are caused by direct action of the cutting elements on the material *i.e.*, cutting tool shape, tool feed rate or by some other disturbances such as friction , wear or corrosion.**

(II) SECONDARY TEXTURE (WAVINESS)

- The surface irregularities of considerable wavelength of a periodic character are called secondary texture or waviness. These irregularities result due to inaccuracies of slides, wear of guides, misalignment of centres, non-linear feed motion, deformation of work under the action of cutting forces, vibrations of any kind etc.

ELEMENTS OF SURFACE TEXTURE



SURFACE

The surface of a part is confined by the boundary which separates that part from another part, substance or space.

ACTUAL SURFACE

This refers to the surface of a part which is actually obtained after a manufacturing process.

NOMINAL SURFACE

A nominal surface is a theoretical, geometrically perfect surface which does not exist in practice, but it is an average of the irregularities that are superimposed on it.

PROFILE

- It is the contour of any section through a surface

ROUGHNESS

Roughness refers to relatively finely spaced microgeometrical irregularities. It is also called as primary texture and constitutes third and fourth order irregularities.

ROUGHNESS HEIGHT

This is rated as the arithmetical average deviation expressed in micrometers normal to an imaginary centre line, running through the roughness profile

ROUGHNESS WIDTH

Roughness width is the distance parallel to- the normal surface between successive peaks or ridges that constitutes the predominant pattern of the roughness.

ROUGHNESS WIDTH CUTOFF

This is the maximum width of surface irregularities that is included in the measurement of roughness height. This is always greater than roughness width and is rated in centimetres.

WAVINESS

Waviness consists of those surface irregularities which are of greater spacing than roughness and it occurs in the form of waves. These are also termed as macro geometrical errors and constitute irregularities of first and second order. These are caused due to misalignment of centres, vibrations, machine or work deflections, warping. etc.

EFFECTIVE PROFILE

It is the real contour of a surface obtained by using instrument

FLAWS

Flaws are surface irregularities or imperfections which occur at infrequent intervals and at random intervals. Examples are: scratches, holes, cracks, porosity etc. These may be observed directly with the aid of penetrating dye or other material which makes them visible for examination and evaluation.

SURFACE TEXTURE

Repetitive or random deviations from the nominal surface which forms the pattern on the surface. Surface texture includes roughness, waviness, lays and flaws.

LAY

It is the direction of predominant surface pattern produced by tool marks or scratches. It is determined by the method of production used.

|| = Lay parallel to the boundary line of the nominal surface that is, lay parallel to the line representing surface to which the symbol is applied *e.g.*, parallel shaping, end view of turning and O.D. grinding.

⊥ = Lay perpendicular to the boundary line of the nominal surface, that is lay perpendicular to the line representing surface to which the symbol is applied, *e.g.*, end view of shaping, longitudinal view of turning and O.D. grinding.

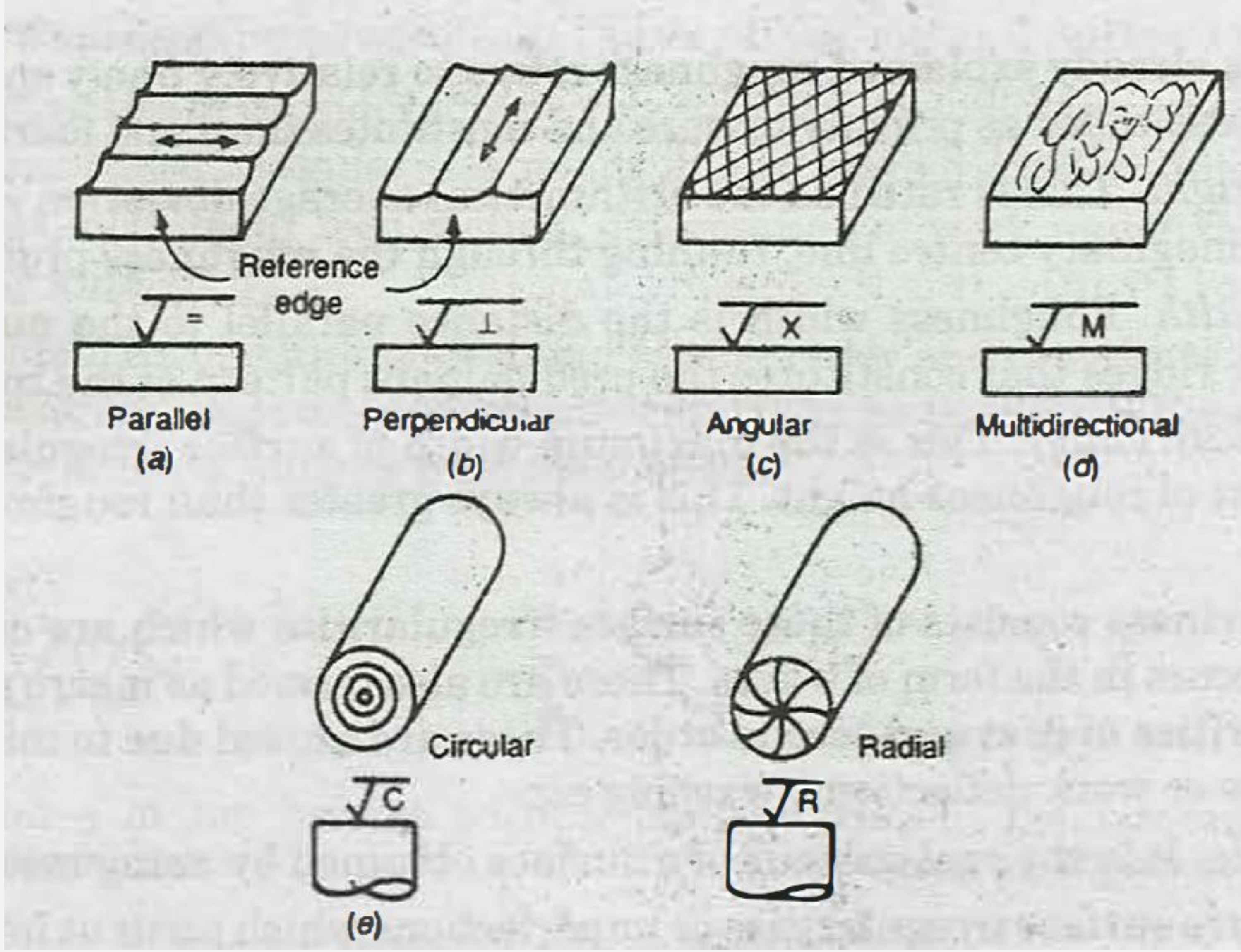
X = lay angular in both directions to the line representing the surface to which symbol is applied, *e.g.*, traversed end mill, side wheel grinding.

M = Lay multidirectional *e.g.* lapping, super finishing, honing.

C = Lay approximately circular relative to the centre of the surface to which the symbol is applied *e.g.*, facing on a lathe.

R = Lay approximately radial relative to the centre of the surface to which the symbol is applied, *e.g.*, surface ground on a turntable, fly cut and indexed on end mill.

Types of lay


Reference
edge
=
⊥
X
M
Parallel
(a)
Perpendicuiar
(b)
Angular
(c)
Multidirectional
(d)
Circular
C
Radial
R
(e)

SAMPLING LENGTH

- It is the length of the profile necessary for the evaluation of the irregularities to be taken into account. It is also known as 'cut-off length. It is measured in a direction parallel to the general direction of the profile. The sampling length should bear some relation to the type of profile. It is found that the required length can be related to the process employed for finishing and a series of Indian Standard Sampling Lengths, or cut-off values, have been evolved to cover the majority of finishing processes normally used. These standard lengths are 0.08, 0.25, 0.8, 2.5 and , 25 mm.

FACTORS AFFECTING SURFACE ROUGHNESS

- W/P material
- Vibrations
- Machining type
- Tool and fixtures

THE GEOMETRIC IRREGULARITIES CAN BE CLASSIFIED AS

- 1^{st} order
- 2^{nd} order
- 3^{rd} order
- 4^{th} order

1ST ORDER IRREGULARITIES

- They are caused by the lack of straightness of guide ways on which tool move

2^{ND} ORDER IRREGULARITIES

- They are caused by vibrations

3^{RD} ORDER IRREGULARITIES

- They are caused by machining

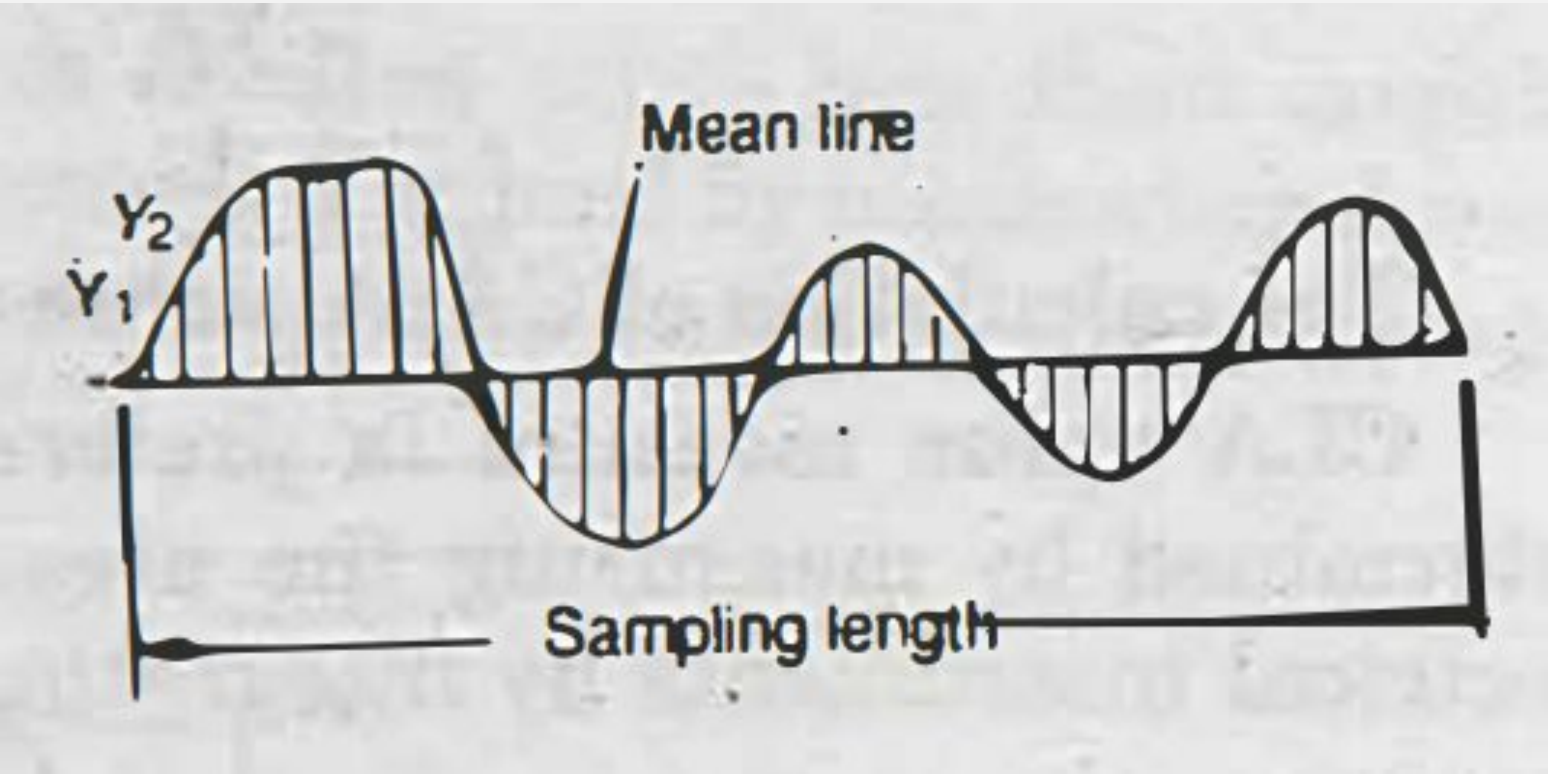
4^{TH} ORDER IRREGULARITIES

- They are caused by improper handling of machines and equipments

MEAN LINE OF PROFILE

It is the line dividing the effective profile such that within the sampling length the sum of the squares of vertical ordinates (y1, Y2, ...) between the effective profile points and the mean line is minimum

MEAN LINE OF PROFILE



CENTRE LINE OF PROFILE

It is the line dividing the effective profile such that the areas embraced by the profile above and below the line are equal. For repetitive wave form (profile) the centre line and mean line are equivalent. Though true repetitive profile is impossible in any manufacturing process, mean line and centre line are assumed to be equivalent for practical purposes.

EVALUATION OF SURFACE FINISH

A numerical assessment of surface finish can be carried out in a number of ways. These numerical values are obtained with respect to a datum. In practice, the following three methods of evaluating primary texture (roughness) of a surface are used

(1) Peak to valley height method

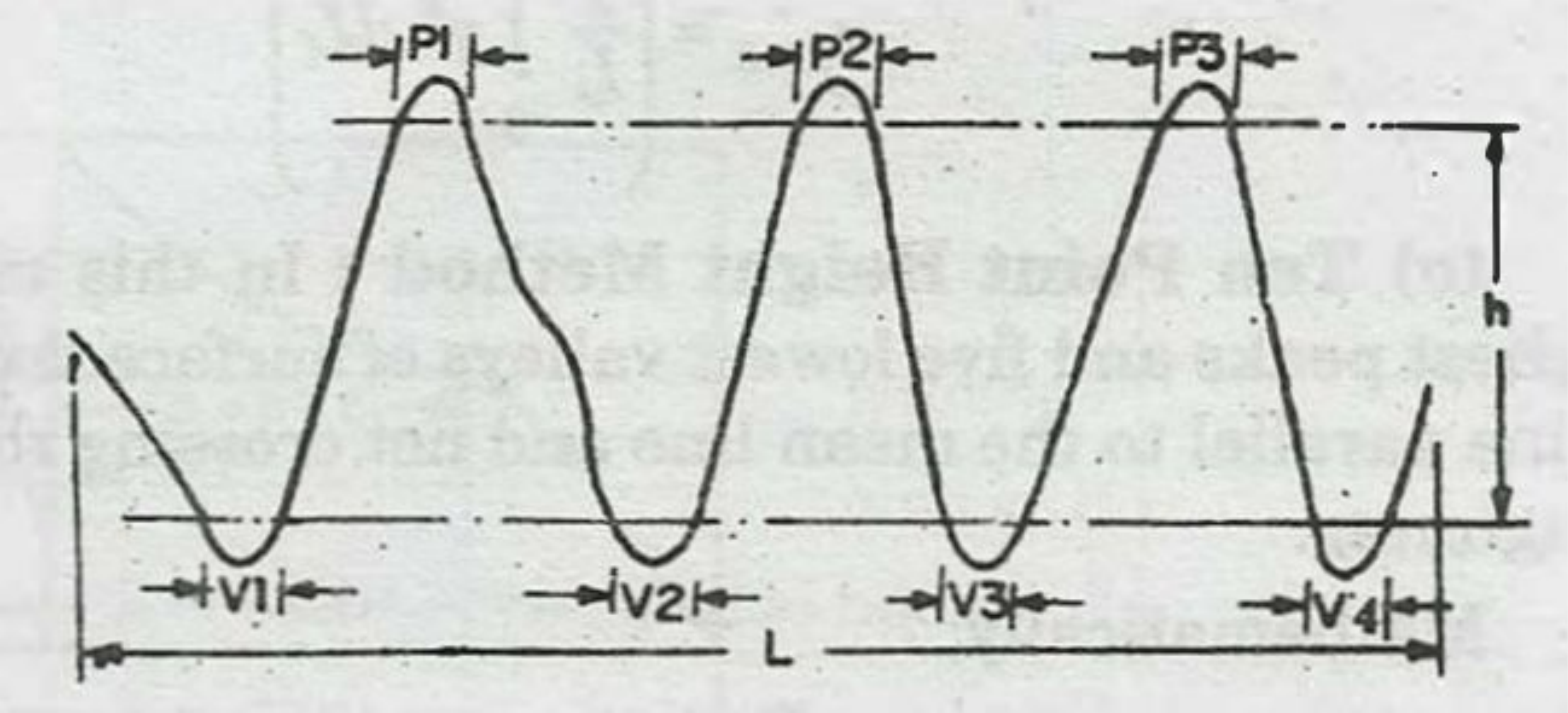
(2) The average roughness

(3) Form factor or bearing curve.

PEAK TO VALLEY HEIGHT

It measures the maximum depth of the surface irregularities over a given sample length, and largest value of the depth is accepted as a measure of roughness. The drawback of this method is that it may read the same h_{max} for two largely different texture. The value obtained would not give a representative assessment of the surface.

PEAK TO VALLEY HEIGHT



PEAK TO VALLEY HEIGHT

To overcome this PV (Peak to Valley) height is defined as the distance between a pair of lines running parallel to the general 'lay' of the trace positioned so that the length lying within the peaks at the top is 5% of the trace length, and that within the valleys at the bottom is 10% of the trace length.

AVERAGE ROUGHNESS

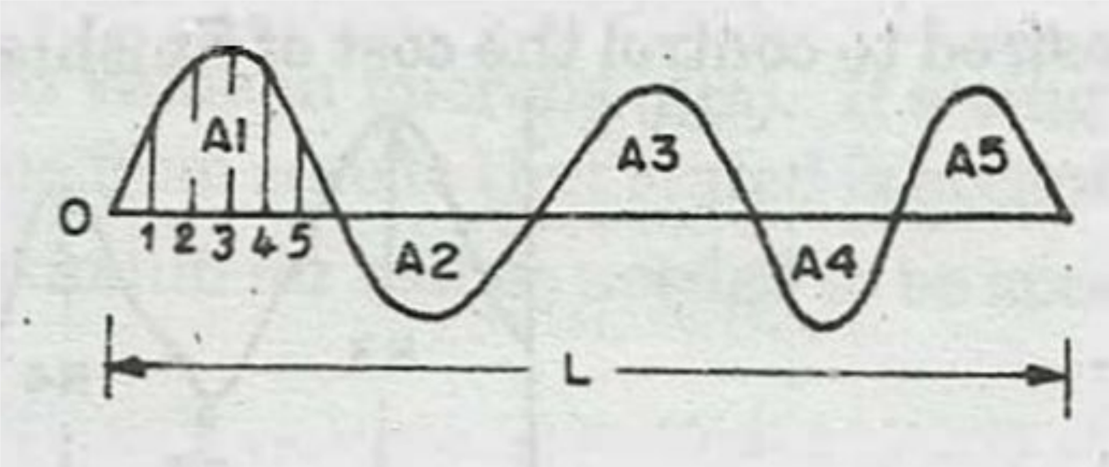
(a) C.L.A. Method

(b) R.M.S. Method

(C) Ten Point Height Method

C.L.A. METHOD

In this method, the surface roughness is measured as the average deviation from the nominal surface. Centre Line Average or Arithmetic Average is defined as the average values of the ordinates from the mean line, regardless of the arithmetic signs of the ordinates.



$$\text{C.L.A Value} = \frac{h_1 + h_2 + h_3 + \ldots\ldots h_n}{n}$$

$$\text{C.L.A.} = \frac{A_1 + A_2 + A_3 + \ldots\ldots A_n}{L}$$

$$= \frac{\Sigma A}{L}$$

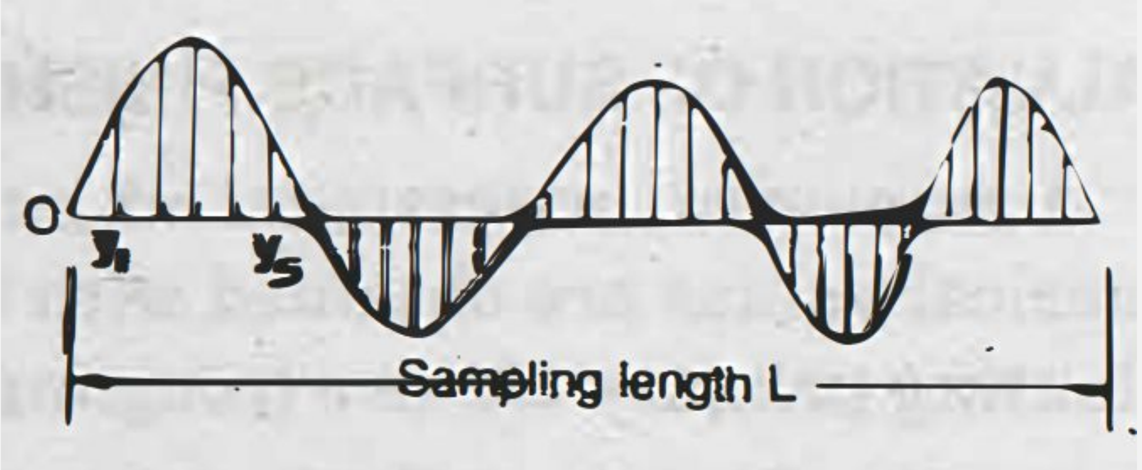
R.M.S. METHOD

In this method also, the roughness is measured as the average deviation from the nominal surface. Root mean square value measured is bas.ed on the least squares.

R.M.S value is defined as the squre root of the arithmetic mean of the values of the squares of the ordinates of the surface measured from a mean line. It is obtained by setting many equidistant ordinates on the mean line (y1, y2, y3, •••) and then taking the root of the mean of the squared ordinates.

R.M.S. METHOD

Let us assume that the sample length 'L' is divided into 'n' equal parts and y1 , y2, Y3, ⋯Yn are the heights of the ordinates erected at those points.



Then, RMS average $= \sqrt{\dfrac{y_1^2 + y_2^2 + y_3^2 + \ldots + y_n^2}{n}}$

or
$$y_{rms} = \left(\frac{1}{L} \int_0^L y^2 \, dL \right)^{1/2}$$

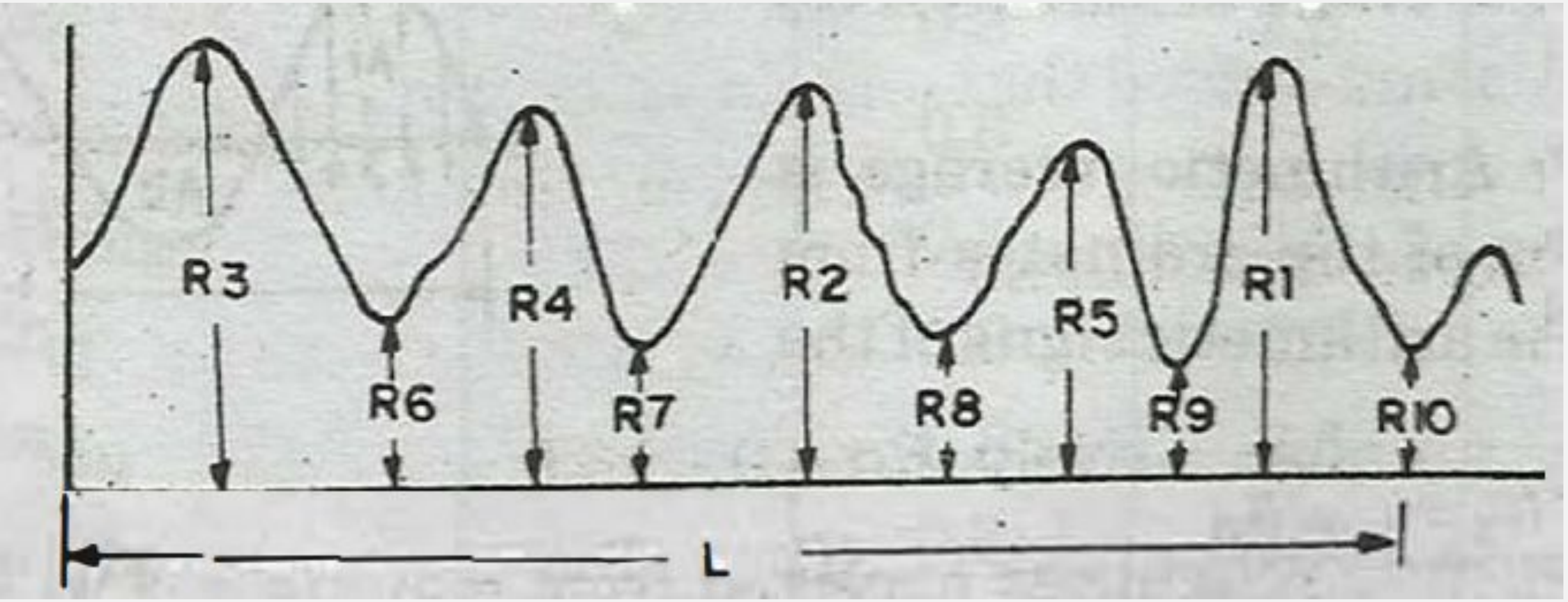
TEN POINT HEIGHT METHOD

In this method, the average difference between the five highest peaks and five lowest valleys of surface texture within the sampling length, measured from a line parallel to the mean line and not crossing the prqfile is used to denote the amount of surface roughness.

TEN POINT HEIGHT METHOD

R_2 = ten point height of irregularities

$$= \frac{1}{5}\left[(R_1 + R_2 + R_3 + R_4 + R_5) - (R_6 + R_7 + R_8 + R_9 + R_{10})\right]$$

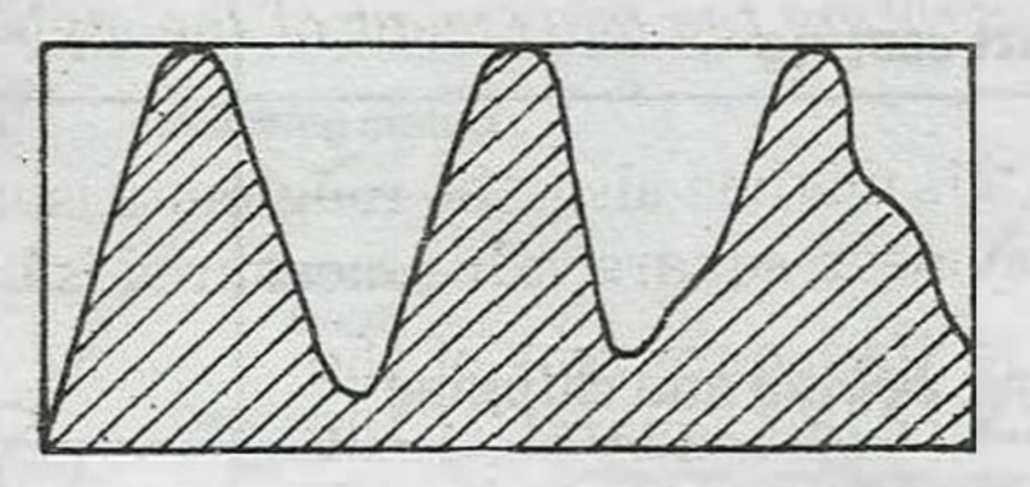


FORM FACTOR AND BEARING CURVES

There are certain characteristics which may be used to evaluate surface texture.

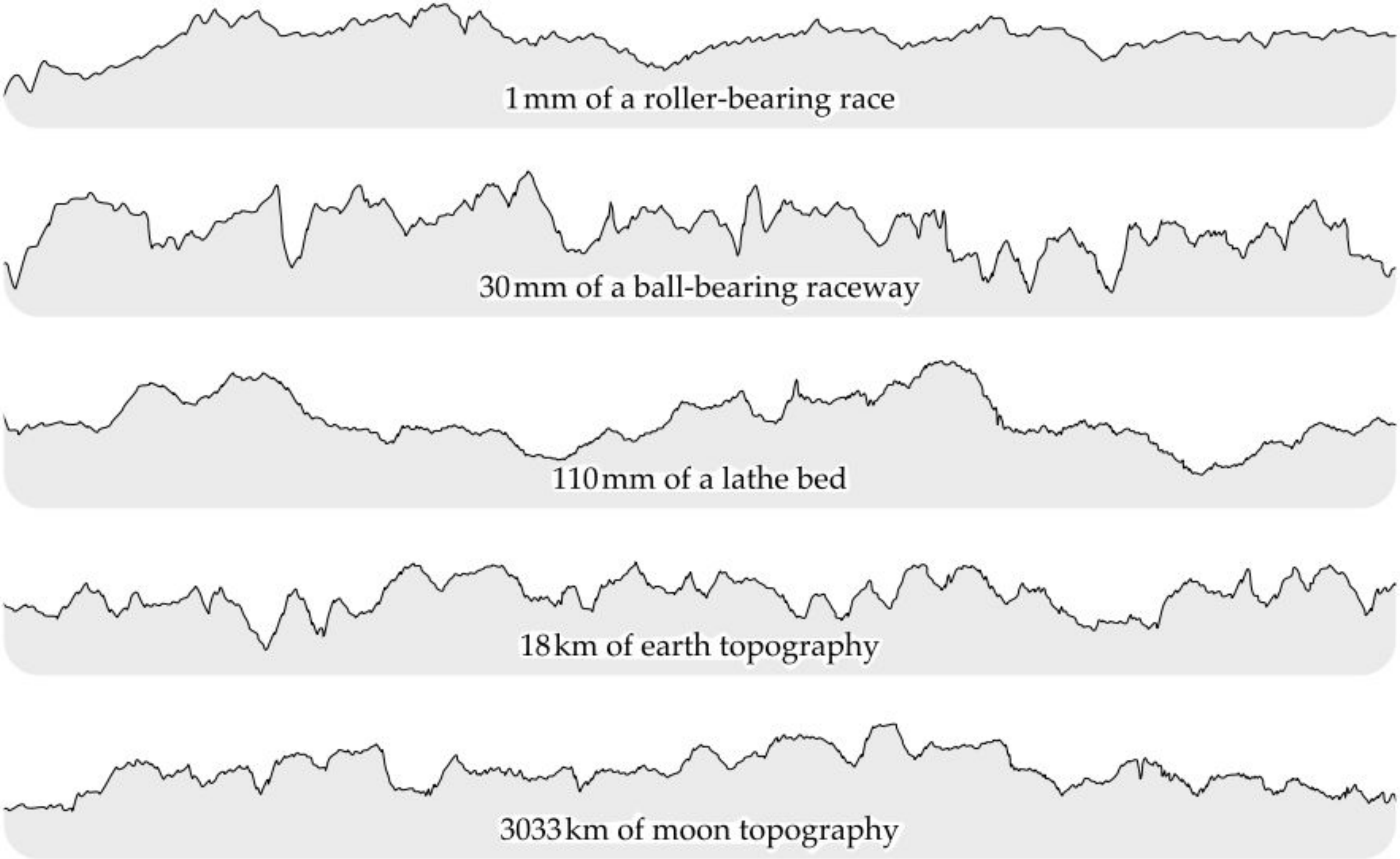
Form Factor : The load carrying area of every surface is often much less than might be thought. This is shown by reference to form factor. The form factor is obtained by measuring the area of material above the arbitrarily chosen base line in the section and the area of the enveloping rectangle.

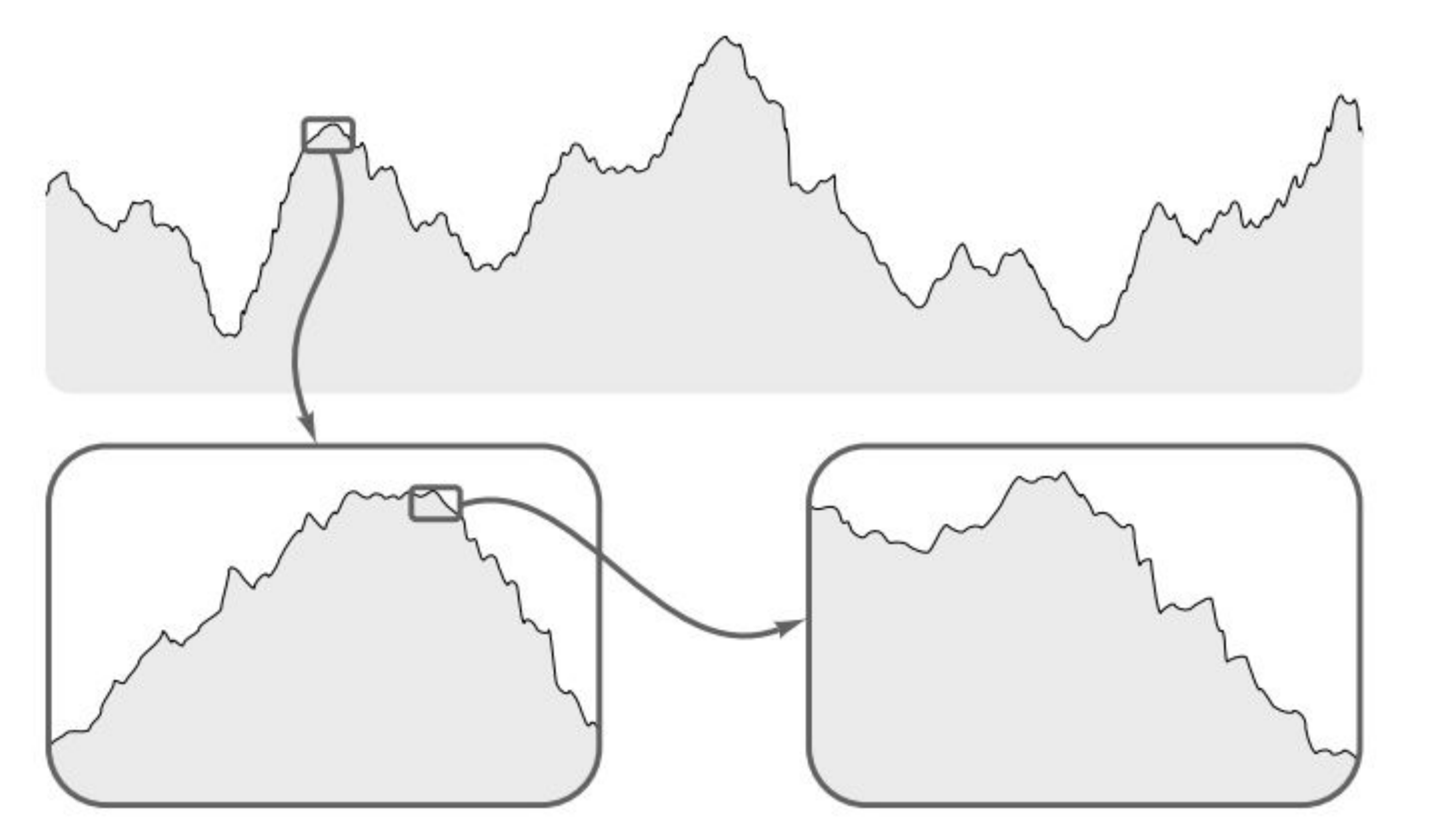
FORM FACTOR



$$\text{Degree of fullness } (K) = \frac{\text{Area of metal}}{\text{Area of enveloping rectangle}}$$

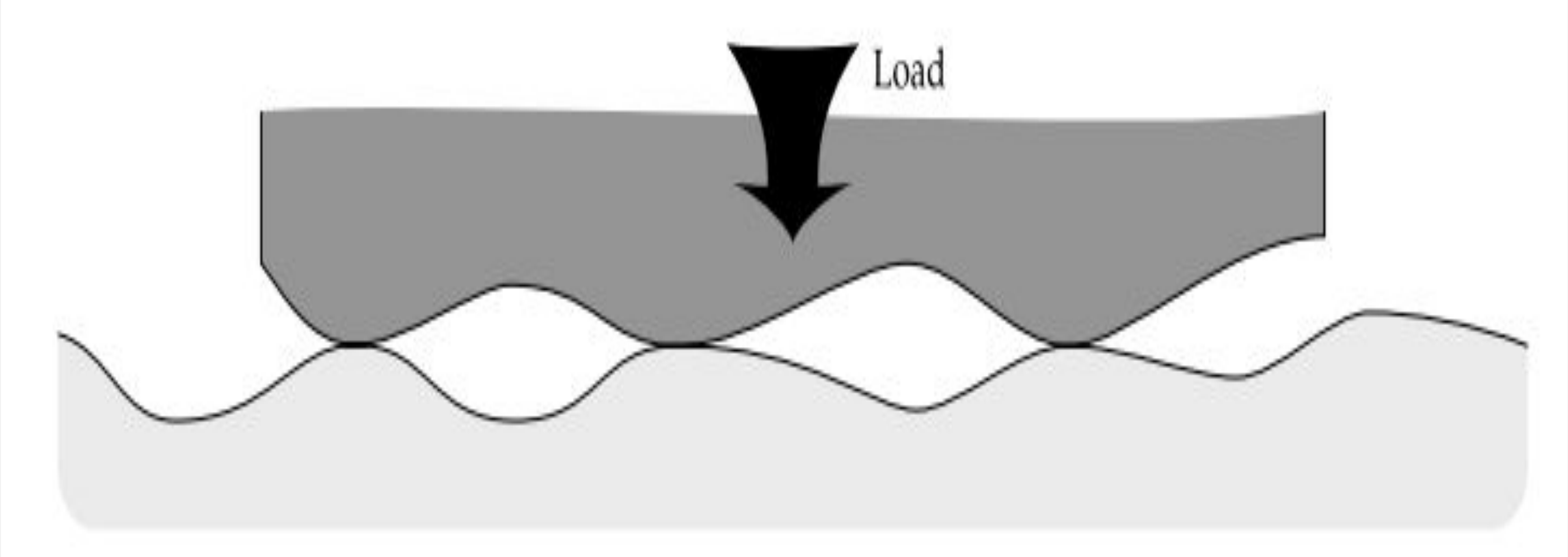
$$\text{Degree of emptiness} = (K_p) = 1 - K$$


1 mm of a roller-bearing race
30 mm of a ball-bearing raceway
110 mm of a lathe bed
18 km of earth topography
3033 km of moon topography

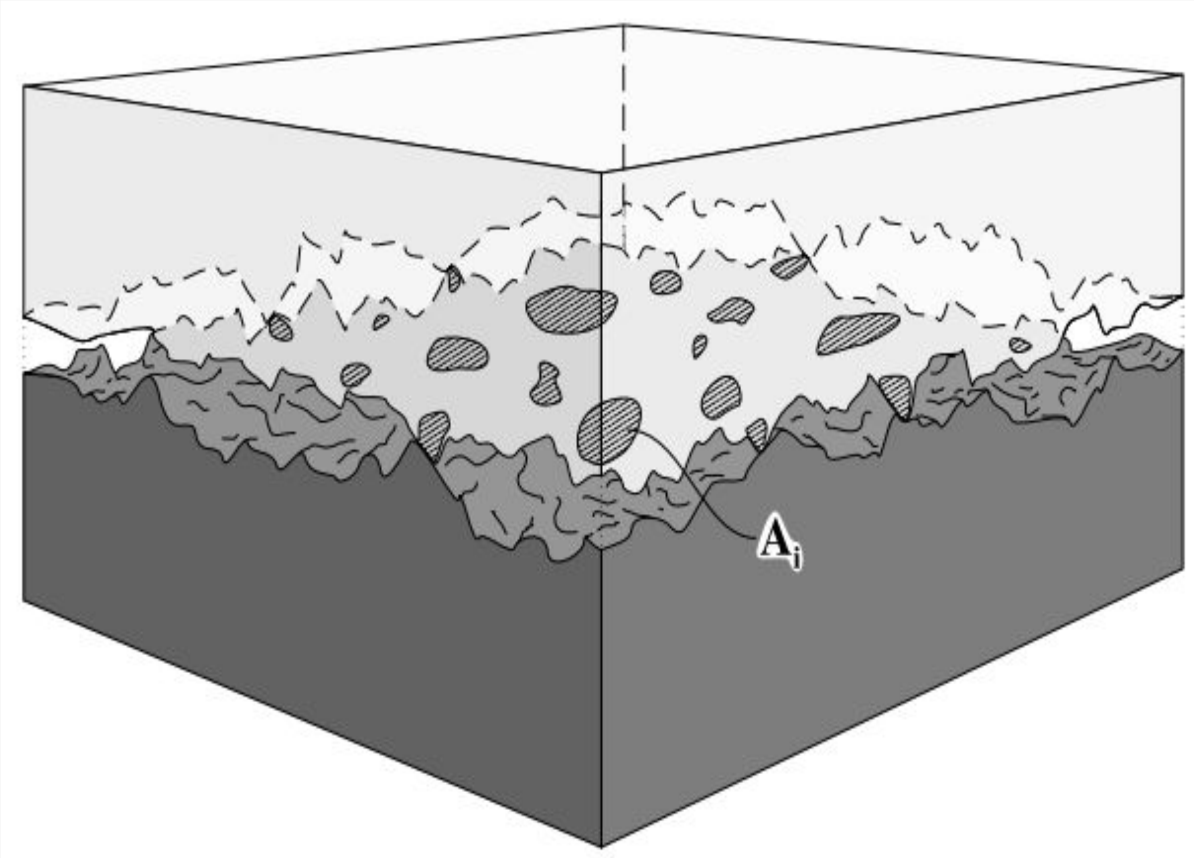


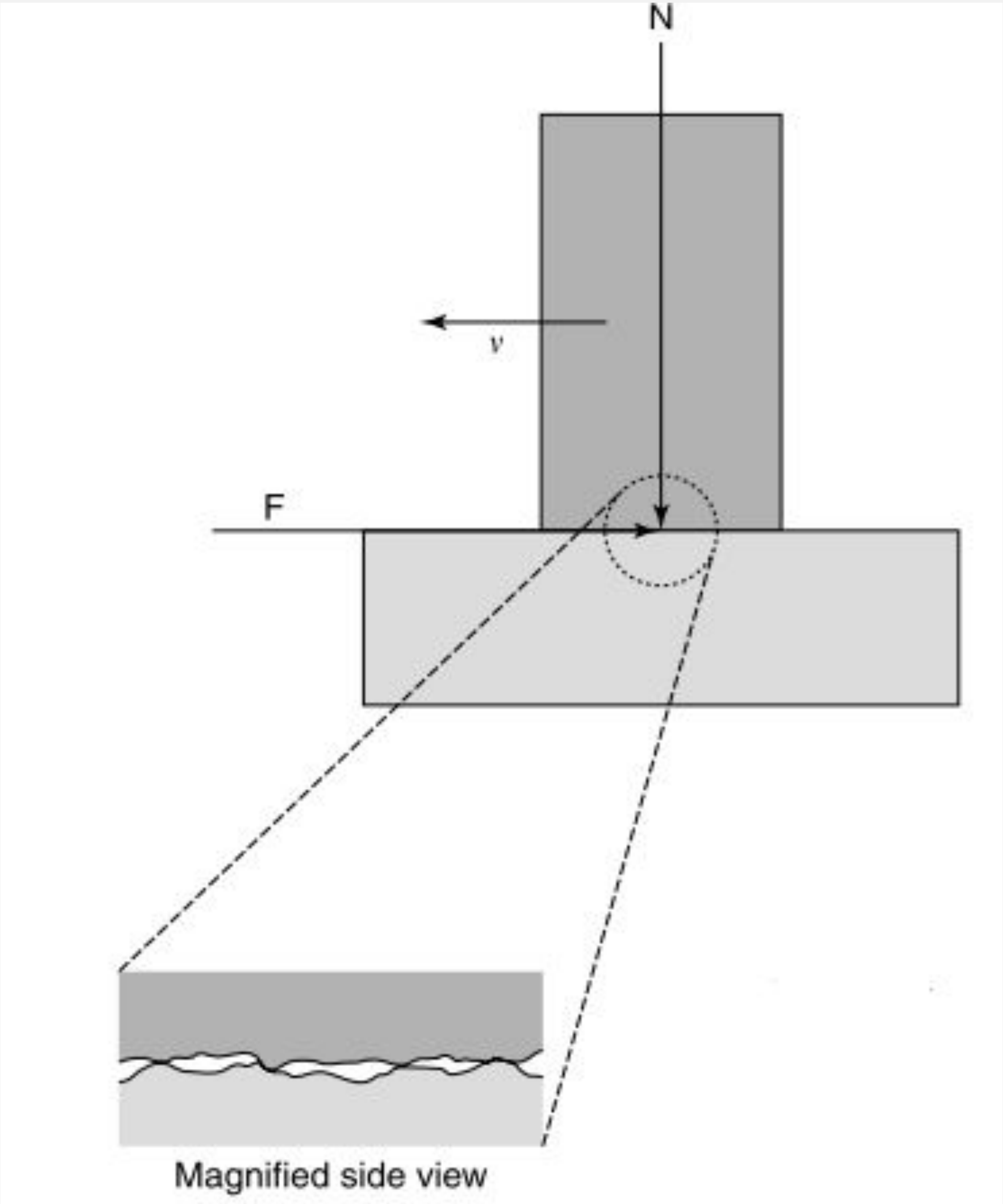
The typical amplitude between peaks and valleys for engineering surfaces is about 1 [μm]

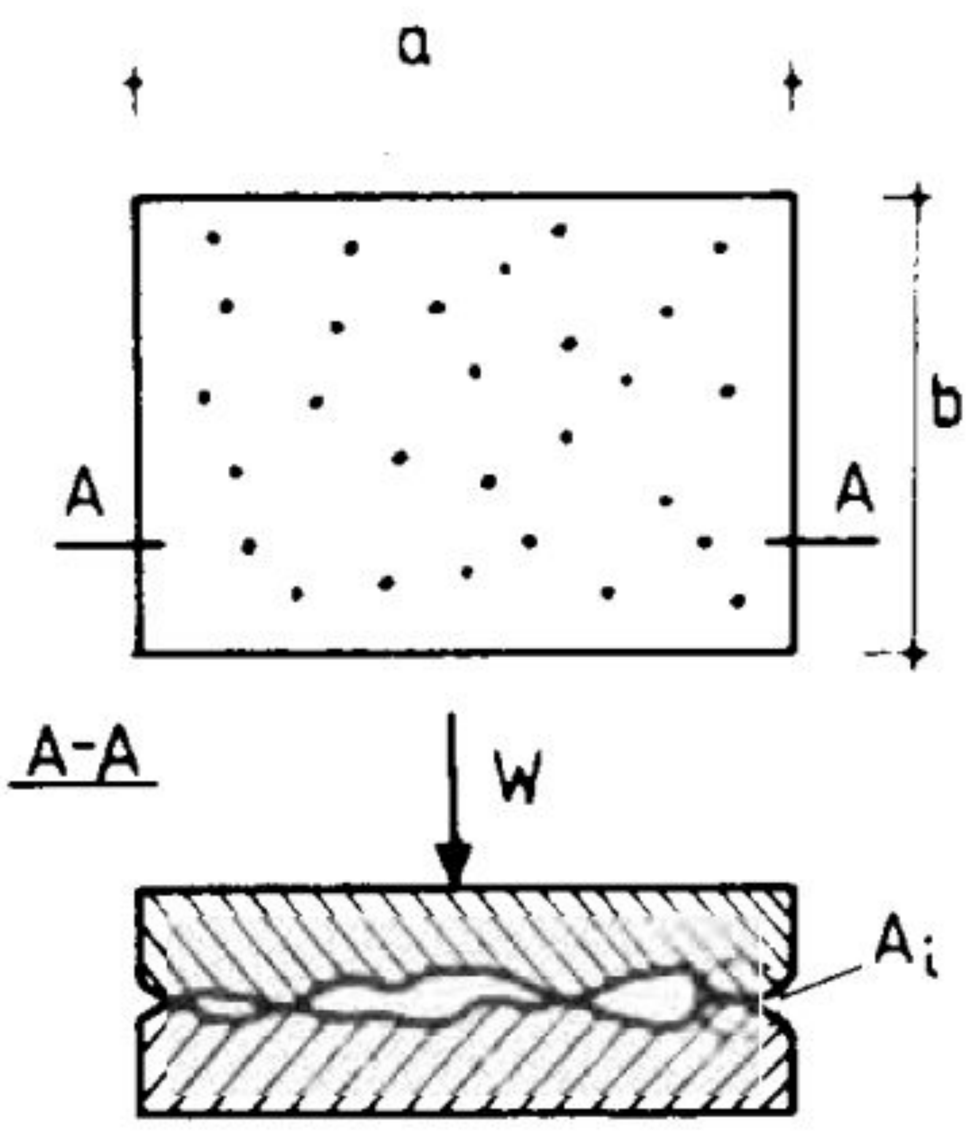
CONTACT BETWEEN **ASPERITIES**



CONTACT BETWEEN SOLIDS




N
v
F
Magnified side view

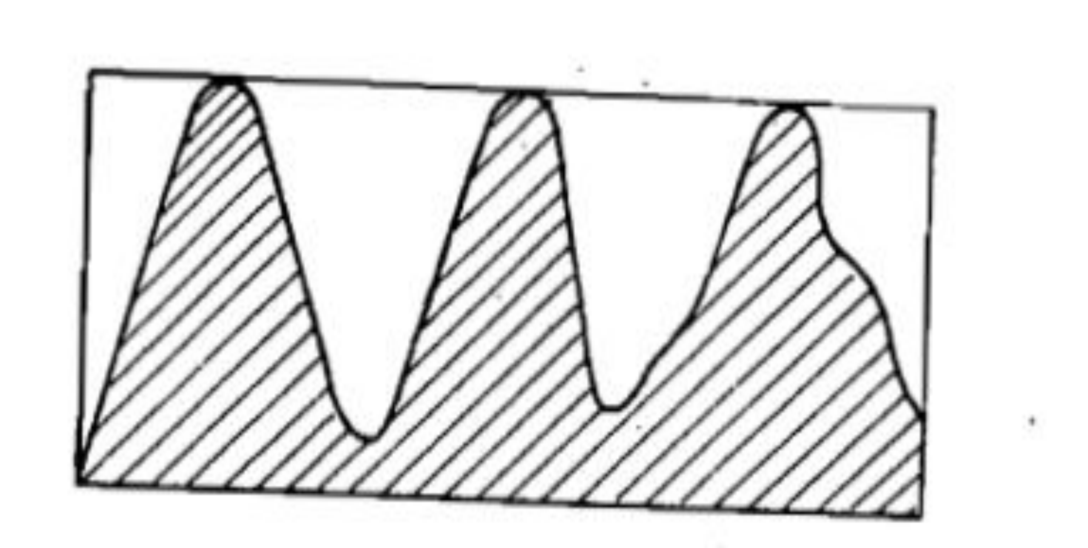


$A_n = a \times b$ (nominal contact area)

$A_r = \sum A_i$ (real contact area)

The load carrying area of every surface is often much less than might be thought. This is shown by reference to form factor. The form factor is obtained by measuring the area of material above the arbitrarily chosen base line in the section and the area of the enveloping rectangle.

FORM FACTOR



$$\text{Degree of fullness } (K) = \frac{\text{Area of metal}}{\text{Area of enveloping rectangle}}$$

$$\text{Degree of emptiness} = (K_p) = 1 - K$$

BEARING AREA CURVE OR ABBOT'S CURVE

- This curve representing the real contact area, is obtained from the surface profile. It is compiled by considering the fraction of surface profile intersected by an infinitesimally thin plane positioned above a datum plane. The intersect length with material along the plane is measured, summed together and plotted as a proportion of the total length. The procedure is repeated through a number of slices. The proportion of this sum to the total length of bearing line is considered to represent the proportion of the true area to the nominal area .

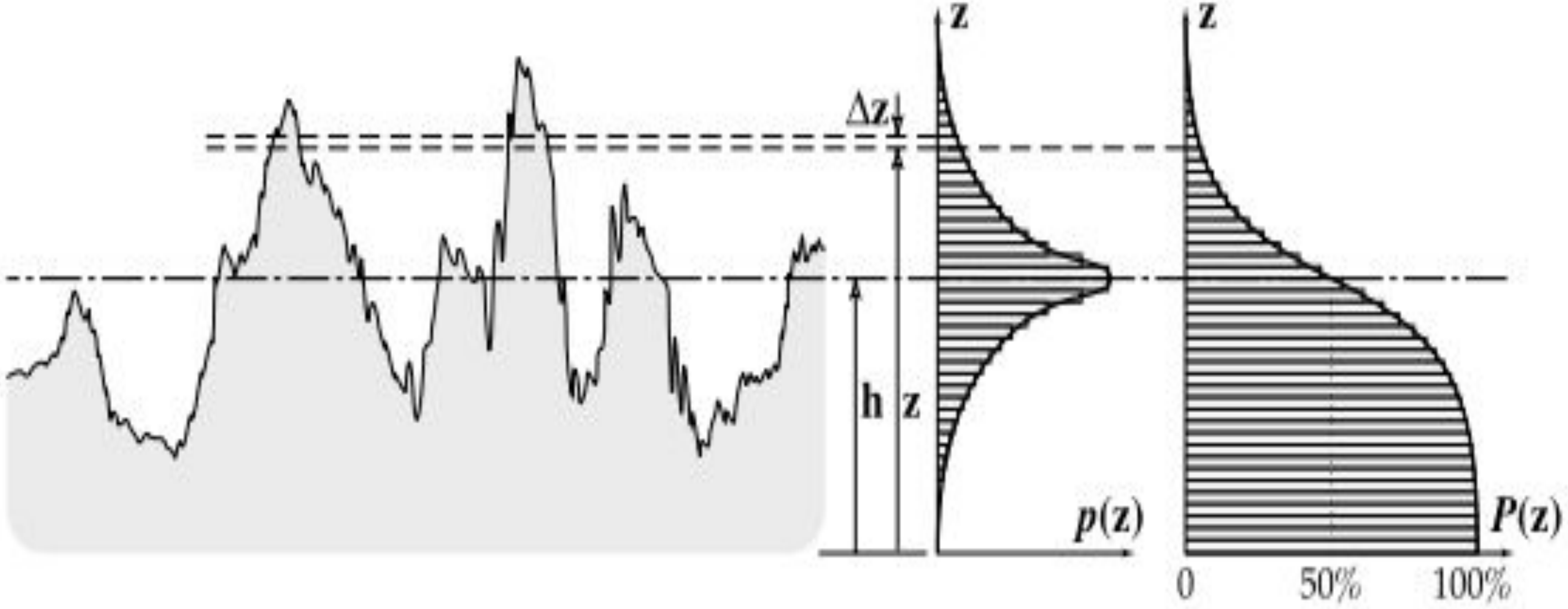
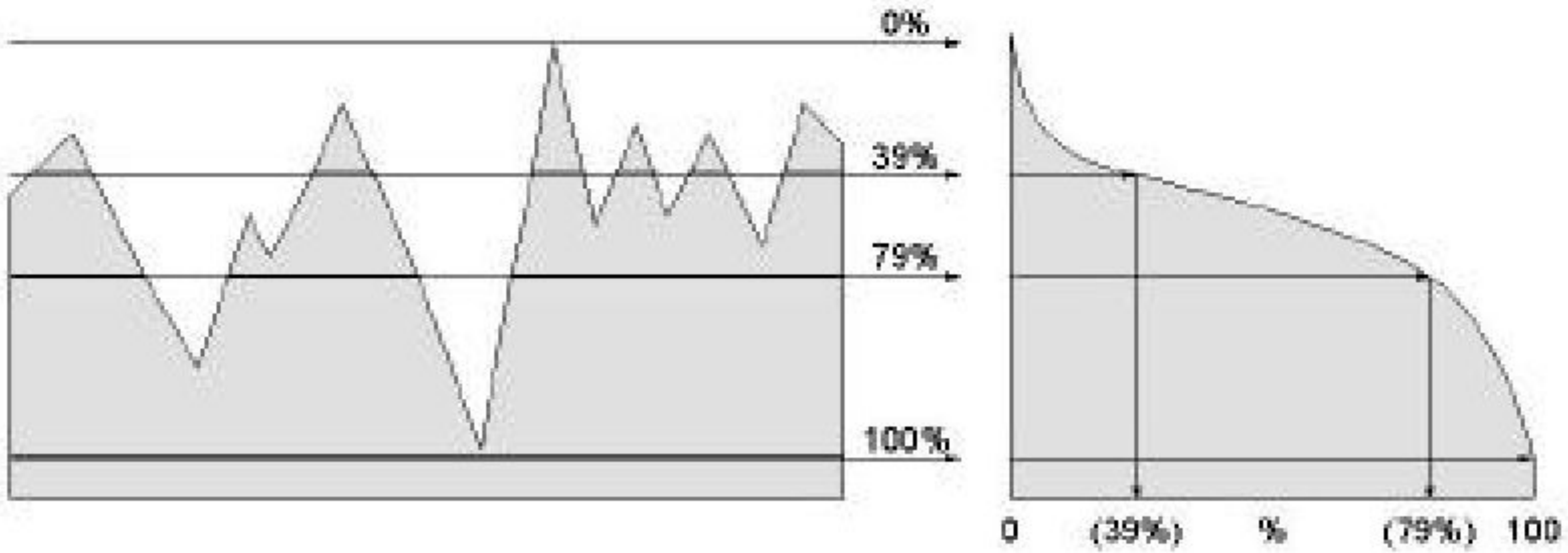
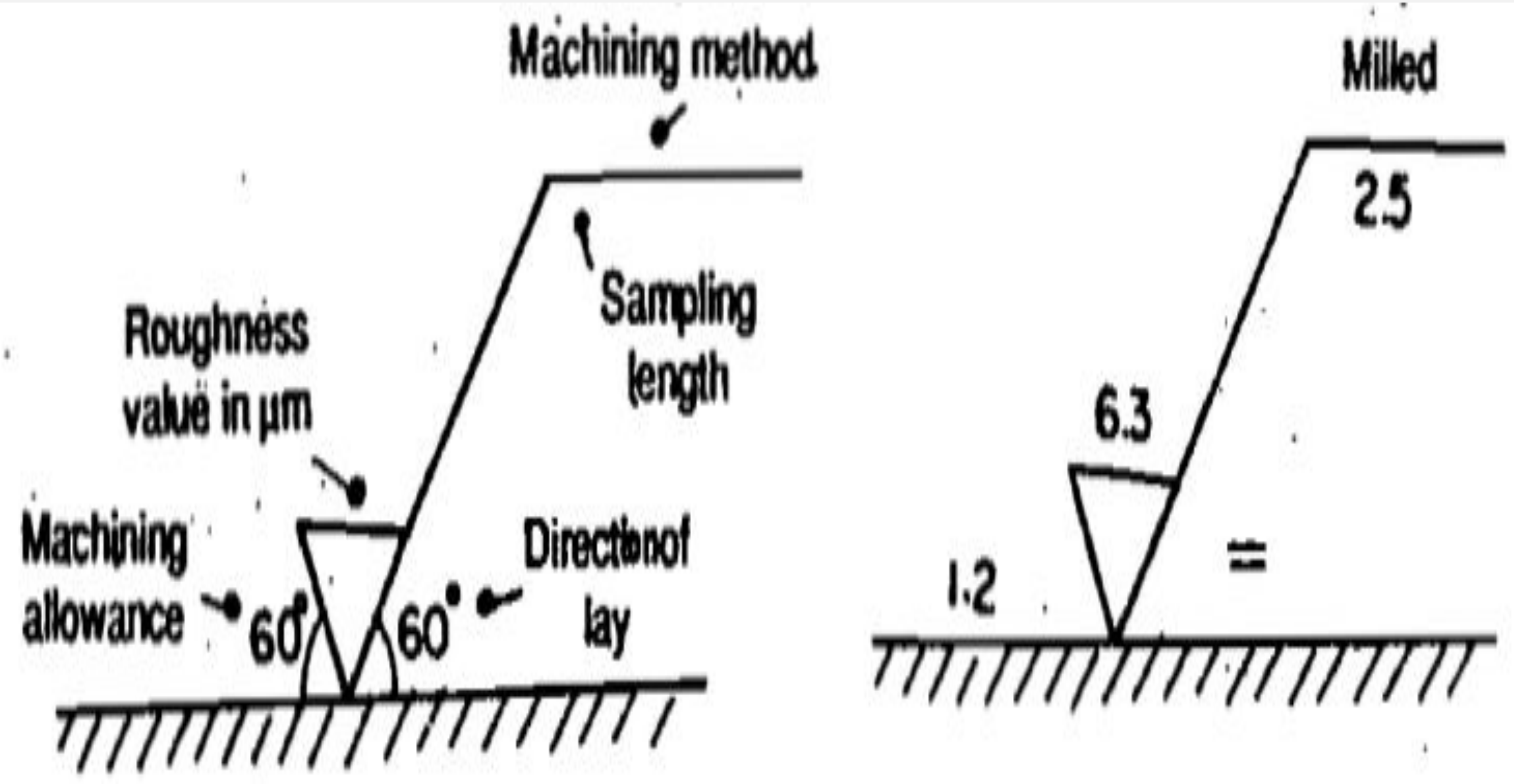


FIGURE 10.4 Determination of a bearing area curve of a rough surface; **z** is the distance perpendicular to the plane of the surface, **Δz** is the interval between two heights, **h** is the mean plane separation, **p(z)** is the height probability density function and **P(z)** is the cumulative probability function [9].

- Bearing Ratio Curve
  - A representation of bearing ratio at various depths of the surface.





Machining method
Sampling length
Roughness value in µm
Machining allowance
60°
60°
Direction of lay
Milled
2.5
6.3
1.2
=

Symbol	*Ra value in μm*
~	above 25
∇	8 to 25
∇∇	1.6 to 8
∇∇∇	0.25 to 16
∇∇∇∇	< 0.025

MEASUREMENT OF SURFACE FINISH

Measurement of surface finish or surface texture :

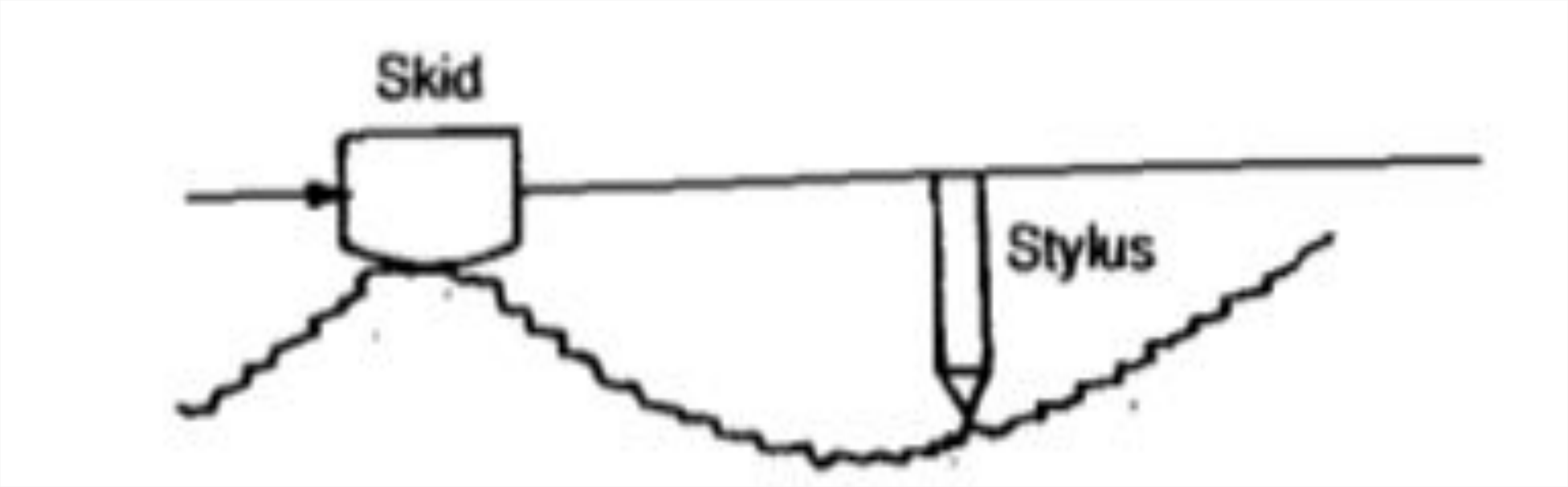
1) Inspection by comparison

2)Direct instrument measurement.

INSPECTION BY COMPARISON

(1) Visual Inspection
(2) Touch Inspection
(3) Scratch Inspection
(4) Microscopic Inspection
(5) Surface photographs
(6) Micro Interferometer
(7) Wallace surface Dynomometer
(8) Reflected Light Intensity

DIRECT INSTRUMENT MEASUREMENT : SKID AND STYLUS INSTRUMENTS



SKID AND STYLUS INSTRUMENTS

Stylus type instruments generally consist of the following units

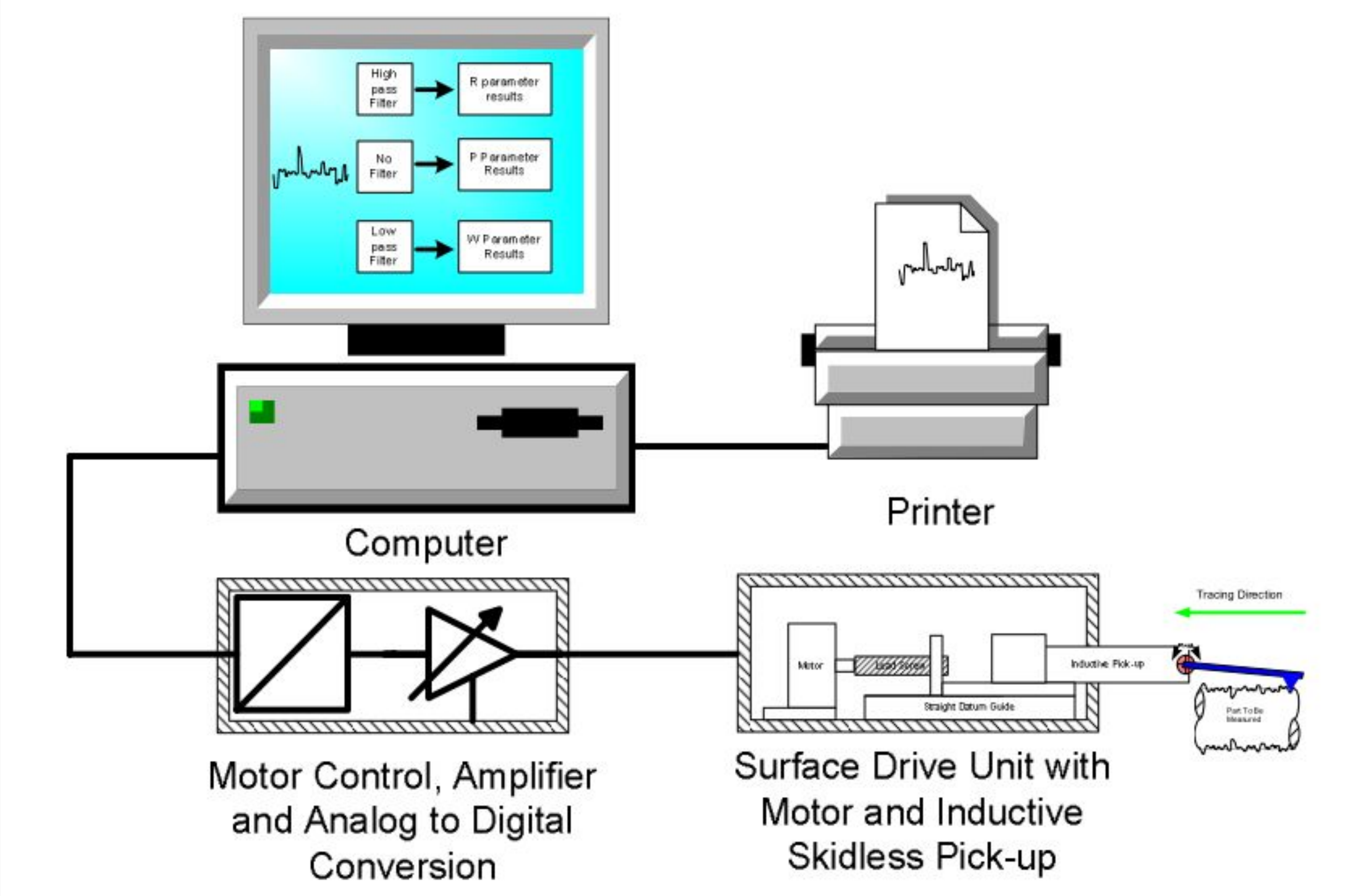
(i) Skid or shoe

(ii) Finely pointed stylus or probe

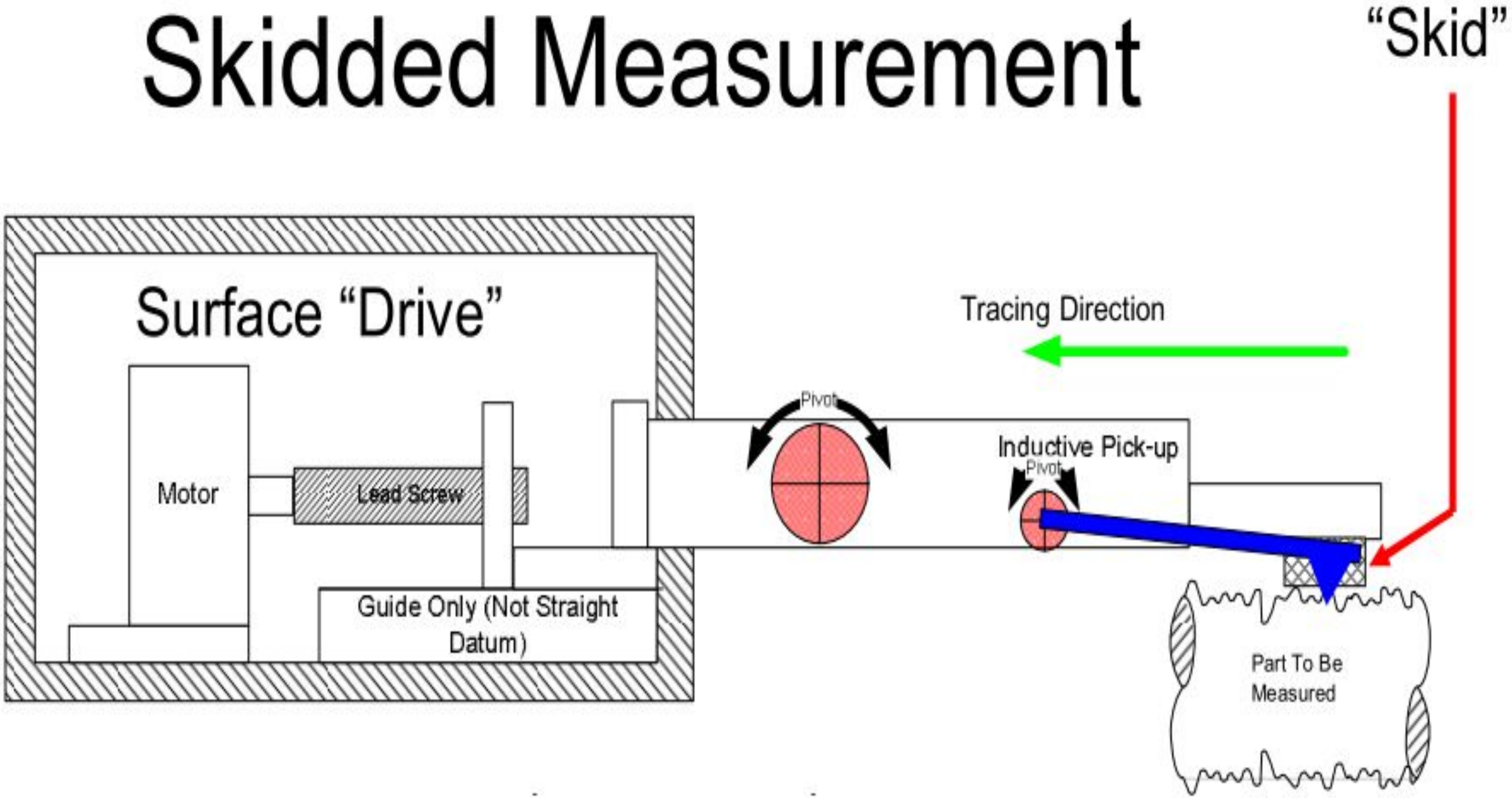
(iii) An amplifying device for magnifying the stylus movement and indicator

(iv) Recording device to produce a trace and

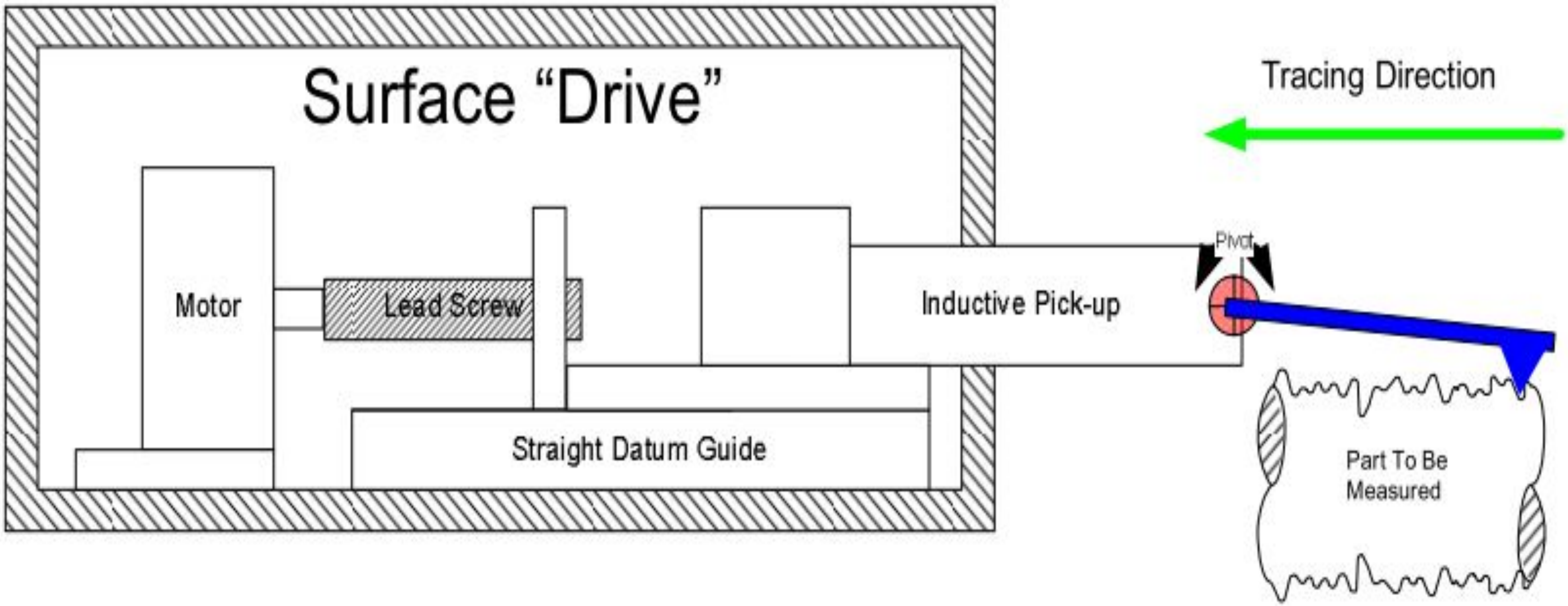
(v) Means for analysing the trace.


High pass Filter
R parameter results
No Filter
P Parameter Results
Low pass Filter
W Parameter Results
Printer
Computer
Tracing Direction
Motor
Inductive Pick-up
Straight Datum Guide
Motor Control, Amplifier and Analog to Digital Conversion
Surface Drive Unit with Motor and Inductive Skidless Pick-up

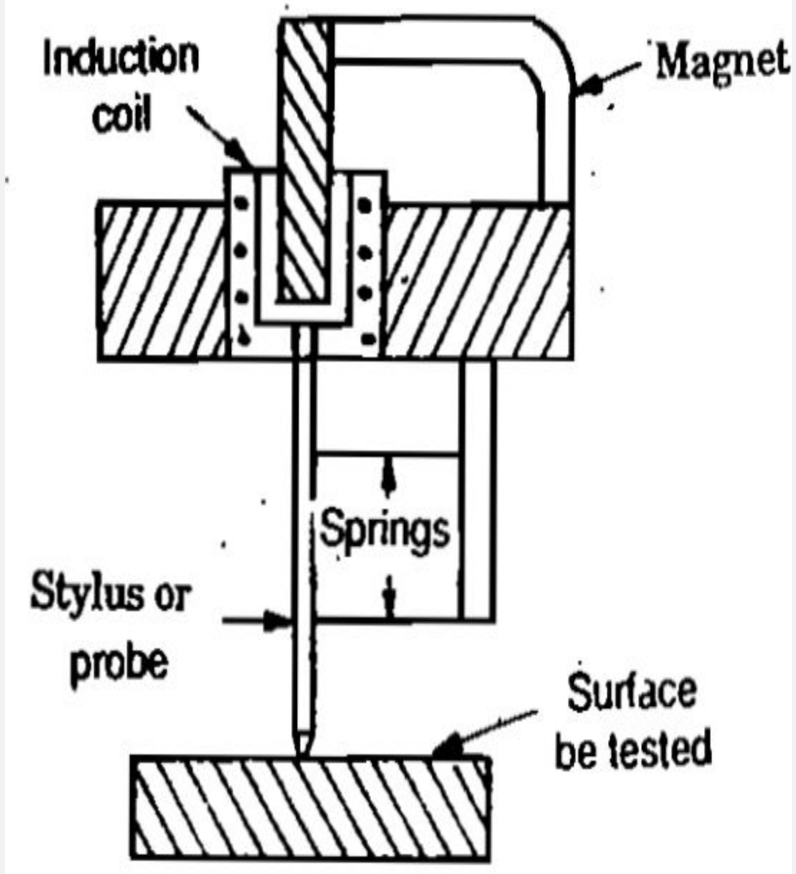
Skidded Measurement



Skidless Measurement

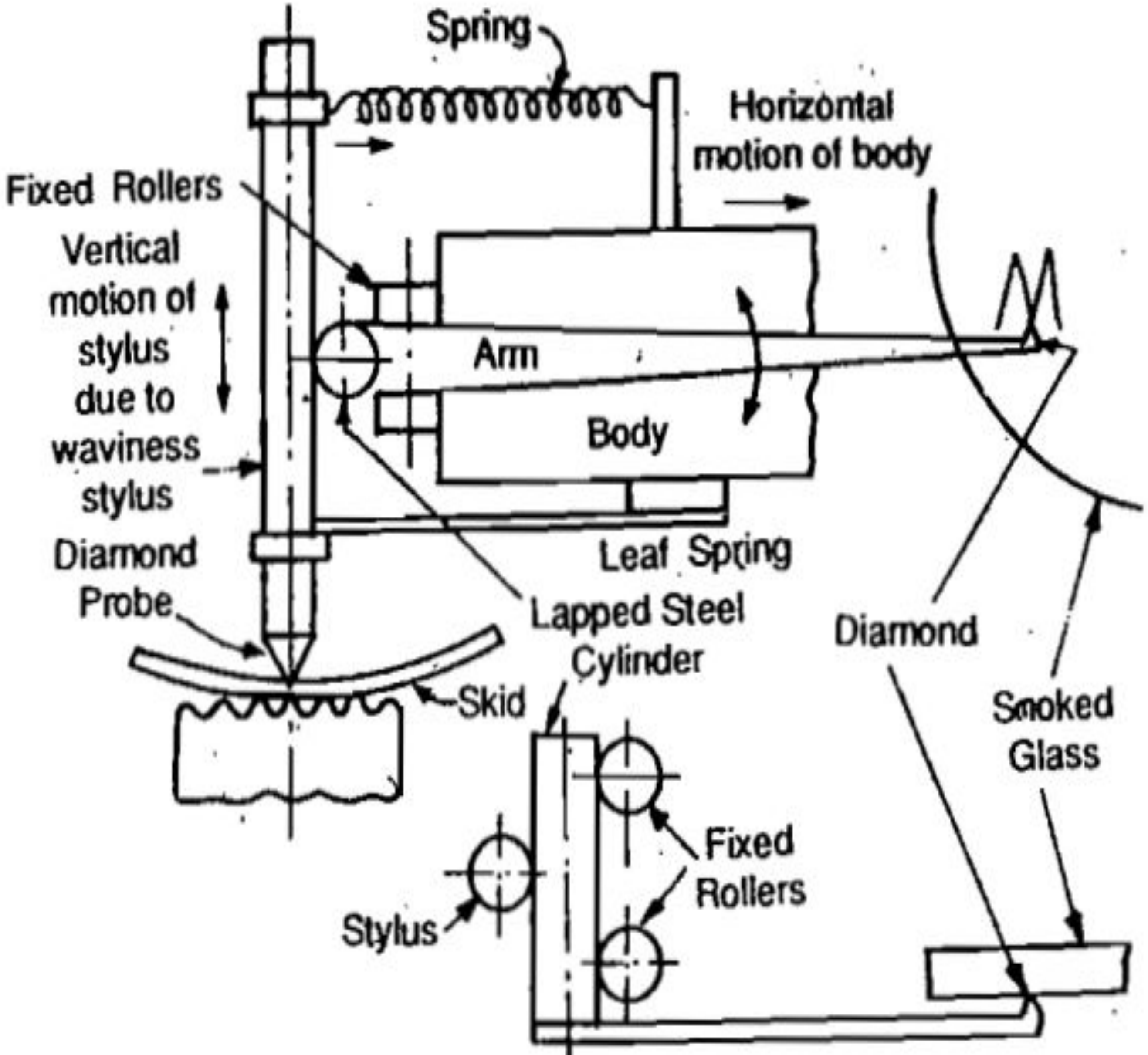


PROFILOMETER



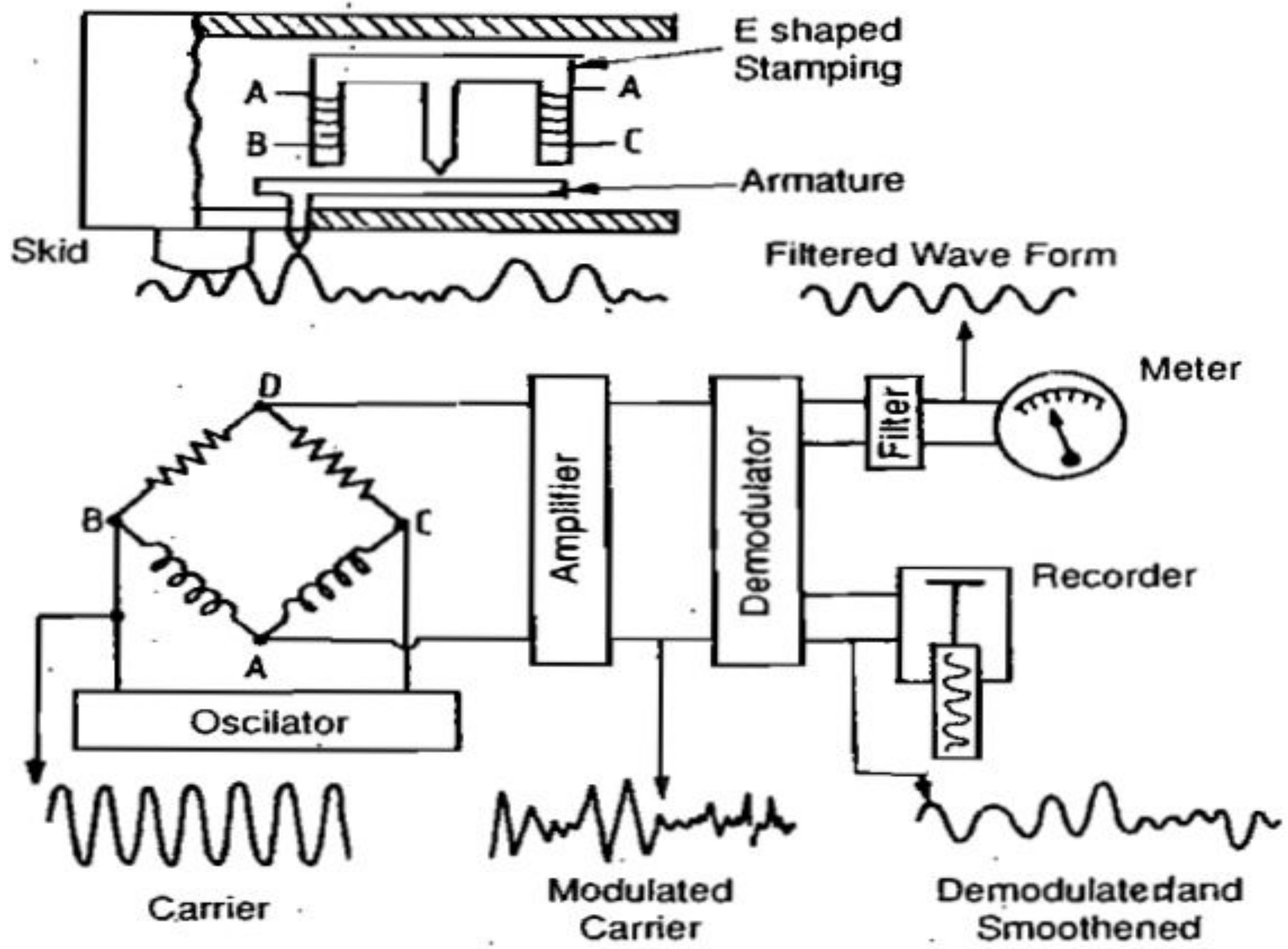
Profilometer

TOMILSON SURFACE METER



Tomlinson Surface Meter

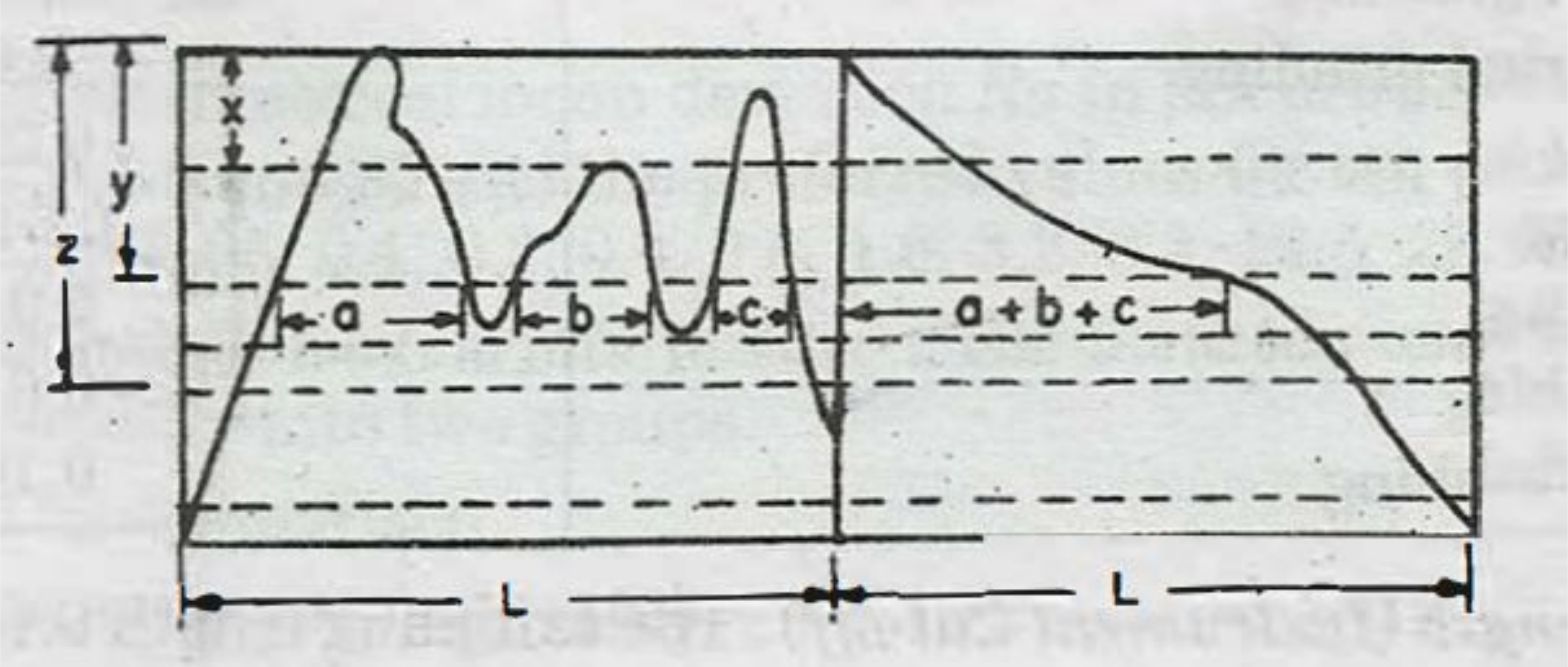
TAYLOR HOBSON TALYSURF


E shaped
Stamping
A
B
A
C
Armature
Skid
Filtered Wave Form
Meter
D
B
C
A
Amplifier
Demodulator
Filter
Recorder
Oscillator
Carrier
Modulated
Carrier
Demodulatedand
Smoothened

BEARING AREA CURVE

The bearing area curve is also called as **Abbot's bearing curve**. This is determined by adding the lengths a, b, c etc. at depths x, y, z etc. below the reference line and indicates the percentage bearing area which becomes available as the crest area worn away.

BEARING AREA CURVE



STATEMENT OF SURFACE ROUGHNESS

(1) Surface Roughness Value : It is expressed as Ra value in microns (μm).

(2) Limiting values : When both minimum and maximum Ra values needed to be specified these shall be expressed as follows

$Ra\ \frac{8.0}{16.0}$ or alternatively, Ra 8.0 – 16.0

(3) Sampling Length (instrument Cut-off): The sampling length is indicated in parenthesis following the roughness value as follows :
Ra 8.0 (2.5)

Here, Ra value is 8.0 microns and sampling length 2.5 mm.

(4) Lay : It is sometimes necessary to specify the direction of lay. It is expressed in accordance with the following example:

Ra 1.5 lay parallel

Unless otherwise specified, it is assumed that the surface roughness shall be across the direction of Lay.

(5) Process : When it is necessary to limit the production of a surface to the use of one particular process, the process shall be stated .

Sr. No.		Manufacturing process	Ra value in μm
1.	Casting	Sand casting	5 to 50
		Permanent mould casting	0.8 to 6.3
		Die casting	0.8 to 3.2
		High pressure casting	0.32 to 2
2.	Hot working	Hot Rolling	2.5 to 50
		Forging	1.6 to 25
		Extrusion	0.16 to 5
		Flame cutting, sawing and chipping	6.3 to 100